SAMPLE

Famous Astronomers: Harlow Shapley

Mapping the Milky Way

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Introduction

Harlow Shapley holds the strange honor of being the astronomer who made the human address permanently wrong, the Missouri newspaper boy who measured the galaxy and found the Sun living in the provinces rather than the capital. Before the man the sky's orthodoxy held the solar system near the center of a modest universe; after the man the Milky Way had swollen into a vast lens of stars with the Sun parked thirty thousand light years from downtown, and every later map of the cosmos being descended from that one displacement. The biography being the series' richest case of the great error made grandly, Shapley's having been wrong about the spiral nebulae and right about the big Galaxy, the two verdicts arriving in the same decade and being argued in the same debate, the man's career being the education of a century by a single scale argument.

The volume follows the library's standard sixteen-room architecture, the sibling biographies sharing the plan: the discovery of the man, the character, the science, the instruments, the features, the contexts, the modern verdicts, the research programs, the discoveries ledger, the mysteries, the futures, the trivia shelf, and the significance verdict closing the case. Readers arriving from the companion biographies will know their way around; first-time guests lose nothing by starting wherever they like.

The book's method being the house style: one unbroken narrative voice, no reference debris, no imported encyclopedia cuttings, just the full account told plainly, sums intact and anecdotes kept within honest bounds. The biography being the spine and the scale being the muscle, the reader being promised the man whole: the crime reporter, the Princeton thesis machine, the mountain measurer, the Harvard builder, the debate's half-loser and the century's full winner, and the closing rooms being the quiet ones where the man's gifts to the modern field are counted.

The load-bearing chronology runs as follows: the birth in Missouri in eighteen eighty-five, the newspaper years at Joplin, the university entry in nineteen three, the Princeton doctorate in nineteen thirteen, the Mount Wilson years from nineteen fourteen, the Galaxy papers of nineteen eighteen, the Great Debate of nineteen twenty, the Harvard directorship from nineteen twenty-one to nineteen fifty-two, the dwarf galaxy discoveries of the thirties, the popular books across the middle decades, the retirement lecturing, and the death in nineteen seventy-two at eighty-seven, those dates anchoring every anecdote that follows, and any reader being welcome to verify them against the sources the final pages name.

Chapter 1: The Discovery of Harlow Shapley

The man's beginning being the least promising in the series' gallery of astronomers, the boy having been born on a farm outside Nashville in Missouri in eighteen eighty-five, the household being a poor one, the schooling having ended for the boy at the fifth grade and the newspaper office having been the next classroom, the sixteen-year-old having been a police and crime reporter for the Joplin dailies, the city room having been the education in the world's rough facts, and the return to school at sixteen having been the first of the self-made man's many self-rescue operations, the story the man loved telling of his being the oldest boy in class by years, the shame having been the fuel.

The university having been chosen by the accident the man retold all his life: the young man's having gone to the University of Missouri to study journalism, the school of journalism's having been founded but not yet opened, the registrar's having pointed the impatient applicant to the next subject down the alphabet, the astronomy course being the letter that came after, the door having been the accident and the walk through it having been pure Shapley, the young man's having found his life's work by the clerk's delay, the story being the biography's own first discovery.

The talent having been immediate: the mathematics having come easily to the reporter's mind, the honors having accumulated, the thesis work on eclipsing binaries having followed, the Princeton years under Russell being the apprenticeship, the doctorate of nineteen thirteen being the union card, the young man's having been the fastest computer of binary orbits of the era, the Martha Betz collaboration's having begun there, the joint method's having become the standard tool, the doctorate being the ticket west, and the discovery of the man being complete when the Mount Wilson invitation arrived: the farm boy having become the instrument the mountain needed.

The mountain years being the discovery in the fullest sense: the nineteen fourteen arrival at the new solar observatory, the sixty-inch being the largest glass then working, the Cepheid program's being the assignment, the variables in clusters being the target, the measuring marathon's having run through the war years, the big Galaxy's

having been assembled from the cluster distances, the Sun's displacement having been the announcement of nineteen eighteen, the man's having been thirty-two when the universe's address was rewritten in his hand, the discovery chapter closing on the arc: the crime desk to the cosmos in fifteen years, the speed of the rise being the story, and the rise's having happened in the decade the galaxies were being argued.

The Harvard move being the discovery's third act: the nineteen twenty-one invitation to direct the Harvard College Observatory, the young measurer's having been handed the oldest institution in American astronomy, the Massachusetts decades being the builder's era, the programs and the people having been expanded, the observatory's having become the field's classroom, the man's having been the director for thirty-one years, the discovery of the institution being the second career, and the chapter's closing being the composite: the reporter, the measurer, and the builder being one man, the trilogy being complete by thirty-six, and the rest of the biography being the long maintenance of a great career rather than its making.

The public man being discovered in the same decade: the Great Debate of nineteen twenty being the celebrity's christening, the scale argument's having been the talk of the field, the magazines having taken the confident Missourian as the new oracle, the man's having been the science celebrity of the twenties, the lectures and the books having followed, the discovery of the man by the public having been as sudden as the discovery of the galaxy's size, and the close of the chapter arriving at the observation this library keeps returning to: subject and era found each other inside the same half-decade, and neither walked away unchanged.

The chronology deserves a fuller record because the speed was the story: the born-in-Missouri boy having entered the university in nineteen three, the master's having arrived by nineteen five, the Princeton doctorate having been finished by nineteen thirteen at twenty-eight, the mountain having been joined at thirty, the Galaxy's map having been published at thirty-three, the Harvard direction having arrived at thirty-six, the whole career having been made before forty, and the long Harvard decades having been the administration of the made career, the record being the compressed early blaze the biographies all note, the blaze having been the reporter's own pace applied to the sky.

His crowd mattered as much as his talent: from the beginning he moved among the strongest figures of the day, Russell being the Princeton master, Hale having been the mountain's builder and the man's sponsor, the Mount Wilson crew having been the field's vanguard, the Harvard inheritance having included the greatest computing staff in the history of the science, company multiplying capability, and the reporter treating that company as one more beat to cover; knowing the right people was itself a discovery, the rooms being the finest on earth, and the young man covering them like a scoop he had earned.

The closing movement reads like an epilogue: retirement from the director's chair came in nineteen fifty-two, the years having continued as the lecturer and the author, the garden and the guests having continued, the man's having lived to eighty-seven, the death in Colorado having come in nineteen seventy-two, the epilogue being the long harvest: the honors having continued, the interviews having preserved the stories, the memoirs having been dictated late, and the man's having outlived the debate's losers and winners both, the longevity being the last asset, the epilogue having allowed the man to watch the vindications arrive in the modern surveys, and the watching having been the reward the career earned.

Consider the Joplin years once more, because the biographies compress them and the compression hides the making. A sixteen-year-old police reporter in a mining town reads the town the way the later astronomer would read the sky: as a field of variables, each one with a story, the bright ones being the noise and the faint ones being the news. The courtrooms teaching the weighing of testimony, the robbery desk teaching the distrust of the convenient story, the deadline teaching the delivery of the true account on time, the whole curriculum having been the scientific method in a cheaper uniform, and the young Shapley having transferred the curriculum to the stars without noticing the change of subject. The chapter being the series' best proof that the sciences having been invented by the newspapers first, the newspapers having been the observational record of the human sky, and the reporter having become the measurer without ever losing the beat, the beat being the biography's own rhythm, and the rhythm being audible in every paper the man signed.

One story deserves the full length, the man's retelling of the Missouri school examination: the boy of sixteen having presented himself to the county examiner for the teaching certificate, the exam having been passed at the earliest legal age, the country school having been taught by the boy for a term, the salary having paid the university's first fees, the story being the making's own arithmetic, and the biography keeping it because the series loving the self-rescue operations, the rescue having been the man's first, the first having been financed by the second, and the whole career having been downstream of the certificate, the certificate being the paper the telescope was built

on, and the paper being the Missouri county's own contribution to the galactic map, the contribution being the unlikeliest in the sciences' history, and the history being the richer for the county's investment.

The discovery chapter should end on the man's own framing of the life: Shapley having described himself as the product of accidents arranged by a stubborn energy, the alphabetical door being the accident, the walking through being the energy, the framing being the Missouri philosophy in one line, and the biography adopting the line as the chapter's last exhibit because the line being the fairest summary the subject ever offered of himself, the self-summary being rare in the series' gallery, most subjects having been less honest or less funny about the luck, and the luck being the better half of every great career, the exhibit being the luck acknowledged, and the acknowledgment being the man's own best characteristic displayed at the threshold of the book.

A closing story completes the discovery chapter: the man's account of the mountain's night when the cluster numbers finally cohered, the distances having been plotted, the lens having appeared in the numbers, and the Sun having fallen to the edge of the plot, the moment being the biography's own epiphany scene, the scene having been retold by the man with the polish the shelf will meet again, the chapter keeping the scene because the series loving the moment the map coheres, the cohering being the science's own drama, the drama being rare enough to be honored wherever it is recorded, and the record being the man's own testimony, the testimony being the last exhibit of the discovery, the discovery being both the man's and the century's, and the century having never quite thanked him properly until the satellites did, the thanks being the reader's own inheritance, the inheritance being the book's whole reason.